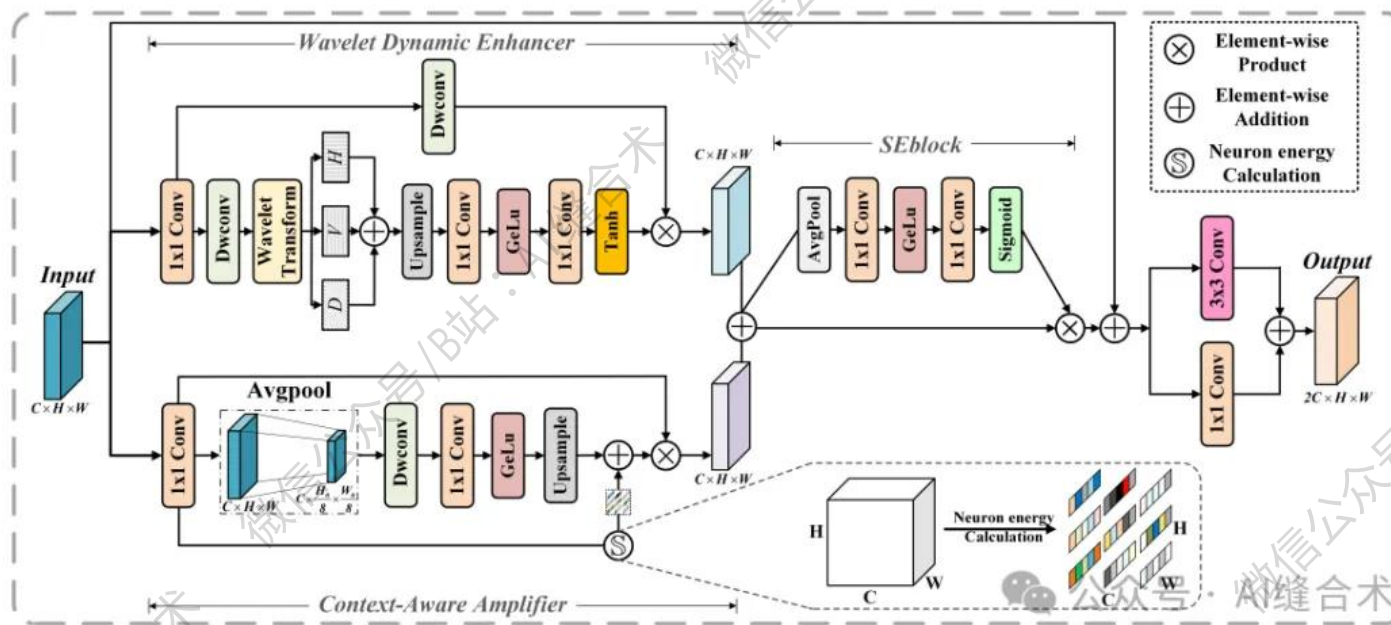


## 核心速览



论文题目：Context-Aware and Semantic-Guided Adaptive Filtering Network for Infrared Small Target Detection

中文题目：用于红外小目标检测的上下文感知和语义引导自适应滤波网络

论文链接：

<https://ieeexplore.ieee.org/document/11305181>

所属单位：重庆大学

核心速览：

本文针对红外小目标检测（ISTD）任务，提出了一种上下文感知与语义引导的自适应滤波网络（CSAFNet）。该网络通过高效建模局部与全局上下文信息，并结合动态特征融合策略，增强目标与背景的区分能力，解决现有混合 CNN-Transformer 方法在特征提取和融合中的局限性。

[https://mp.weixin.qq.com/s/6R4-lxc6b0BtCa2a-ifecQ?token=501365321&lang=zh\\_CN&poc\\_token=HB9aJWqjqppXksheAENEldDhWiMND6wWul3vT9V4](https://mp.weixin.qq.com/s/6R4-lxc6b0BtCa2a-ifecQ?token=501365321&lang=zh_CN&poc_token=HB9aJWqjqppXksheAENEldDhWiMND6wWul3vT9V4)

```
PSAA(
  (LG_att): CAA(
    (conv1_0): Conv2d(32, 64, kernel_size=(1, 1), stride=(1, 1))
    (conv1_1): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1))
    (conv1_2): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1))
    (gelu): GELU(approximate='none')
  )
  (WDE): WDE(
    (dwconv3x3): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32)
    (dwconv3x3_2): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32)
    (conv_0): Down_wt(
      (wt): DWTForward()
      (conv_bn_relu): Conv2d(32, 64, kernel_size=(1, 1), stride=(1, 1))
    )
    (act): GELU(approximate='none')
    (conv_1): Conv2d(64, 32, kernel_size=(1, 1), stride=(1, 1))
  )
  (dwconv): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=32)
  (simam_int): simam_int(lambda=0.000100)
  (ca): Sequential(
    (0): AdaptiveAvgPool2d(output_size=1)
    (1): Conv2d(32, 8, kernel_size=(1, 1), stride=(1, 1))
    (2): GELU(approximate='none')
    (3): Conv2d(8, 32, kernel_size=(1, 1), stride=(1, 1))
    (4): Sigmoid()
  )
)
(Conv3): Sequential(
  (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (2): ReLU(inplace=True)
)
(Conv1): Sequential(
  (0): Conv2d(32, 32, kernel_size=(1, 1), stride=(1, 1))
  (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (2): ReLU(inplace=True)
)
)
input_tensor_shape : torch.Size([2, 32, 128, 128])
output_tensor_shape : torch.Size([2, 32, 128, 128])
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！